Year Six Science Bridging Programme

Project 2
Kitchen Chemistry

Inspire Academic™

Learning How Science Works

Workbook Edition 2026

Inspireacademic.org

INSPIRE ACADEMIC™

YEAR 6 SCIENCE BRIDGING PROGRAMME

PROJECT 2 WORKBOOK

KITCHEN CHEMISTRY

Matter, Mixtures & Change

Welcome, Young Chemist

Chemistry is the science of understanding what things are made of and how they change.

Every day, chemistry is happening all around you. Ice melts, water evaporates, food cooks, metals rust, substances dissolve, and new materials are formed through chemical reactions.

Chemists investigate these changes. They ask questions, carry out experiments, gather evidence, and use what they discover to explain how the world works.

In this project, you will go beyond observing. **You will think, question, and investigate like a real chemist.**

Preface

The INSPIRE Year 6 Science Bridging Programme

The world's best scientists were not born knowing chemistry or physics. They were taught to ask good questions, design fair investigations, gather honest evidence, and explain what that evidence shows. The purpose of this programme is to give every young person, regardless of background or starting point, the habits of mind that make a scientist.

The INSPIRE Year 6 Science Bridging Programme is a carefully sequenced series of project workbooks designed to carry pupils from upper Key Stage 2 into the demands of Key Stage 3 science with confidence. Each project builds directly on the last, so that knowledge, vocabulary, and investigative skills accumulate across the series rather than being treated as isolated units of study.

This workbook is the second in a four-project series. **Project 1: The Great Science Investigator** established the foundations of scientific enquiry: asking good questions, planning fair tests, collecting data, and explaining what evidence shows. If you completed that project, you are already thinking like a scientist. Now, in **Project 2: Kitchen Chemistry**, you will bring those skills to chemistry. You will explore matter, mixtures, reactions, and the acids and alkalis found in your own kitchen.

The series continues with **Project 3: Invisible Worlds**, where pupils investigate electricity, energy, systems, and connections, uncovering the unseen forces that power everyday life. **Project 4: Living Systems** then turns to biology: cells, organs, and the living systems that sustain life. Taken together, the four projects take a pupil from upper Key Stage 2 to the threshold of secondary science. They will arrive with strong knowledge, good scientific vocabulary, and the investigative habits that make science genuinely meaningful.

About This Workbook: Project 2, Kitchen Chemistry

Chemistry is one of the foundational sciences that helps us make sense of the world around us. At primary school, pupils observe many fascinating changes: ice melting, salt dissolving, bicarbonate of soda fizzing with vinegar. At secondary school, they will be expected to explain these observations using particle theory, chemical language, and scientific evidence.

This workbook is designed to bridge that gap. It covers the core content of the KS2 Science Programme of Study, specifically the Year 5 strand on Properties and Changes of Materials. It then builds deliberately towards the KS3 Chemistry curriculum, introducing the vocabulary and ideas pupils will meet from Year 7 onwards.

📖 What This Workbook Covers

Section 1 — Matter & Particles: particle theory, three states of matter, state changes
Section 2 — Mixtures & Separation: dissolving, seven separation methods
Section 3 — Physical & Chemical Change: signs of chemical change, reactants and products
Section 4 — Kitchen Acids & Alkalis: pH scale, indicators, neutralisation
Section 5 — Chemistry Vocabulary: 25 key terms for KS3 readiness
Section 6 — Preparing for Year 7: laboratory skills, safety, what comes next

A Note on Pedagogy

In this workbook, we do not ask pupils to memorise facts. Rather, we ask them to observe carefully, think about particles and change, explain their observations in scientific terms, and build confidence with precise vocabulary.

Investigations sit at the heart of every section. They are the means of learning, not an afterthought. Pupils who complete this project will arrive at secondary school with solid knowledge, confident vocabulary, and the investigative habits that will serve them well across all sciences.

Take your time. Think carefully. Be curious.

By the end of this project, you will understand some of the most important ideas in chemistry.

More than that, you will think like a chemist!

E. K. Appiah

Inspire Academic™

London, 2026

info@inspireacademic.org | inspireacademic.org

PROJECT BIG QUESTION

What is matter made of, and how can we tell when it changes?

Keep this question in mind as you work through this project.

Your 5-Week Learning Journey

Week	Focus	Investigation	What You Produce
Week 1	States of Matter & Particles	Ice, Water, Steam	Particle diagrams
Week 2	Dissolving & Mixtures	The Dissolving Race	Results table & conclusion
Week 3	Separating Mixtures	The Separation Challenge	Separation plan & diagrams
Week 4	Chemical Changes	Fizz, Foam & Gas	Evidence of reaction
Week 5	Kitchen Acids & Alkalis	Kitchen pH Detective	pH results table

What You Will Learn To Do

By the end of Project 2, you will be able to:

- explain what matter is made of using particle theory
- describe the properties of solids, liquids, and gases
- identify and separate mixtures using scientific methods
- distinguish between physical and chemical changes
- observe and record evidence of chemical reactions
- use scientific vocabulary to explain what you observe

SECTION 1

MATTER & PARTICLES

What Is Matter?

Matter is anything that has mass and takes up space.

Everything you can see, touch, and measure is made of matter: water, air, wood, metal, even you!

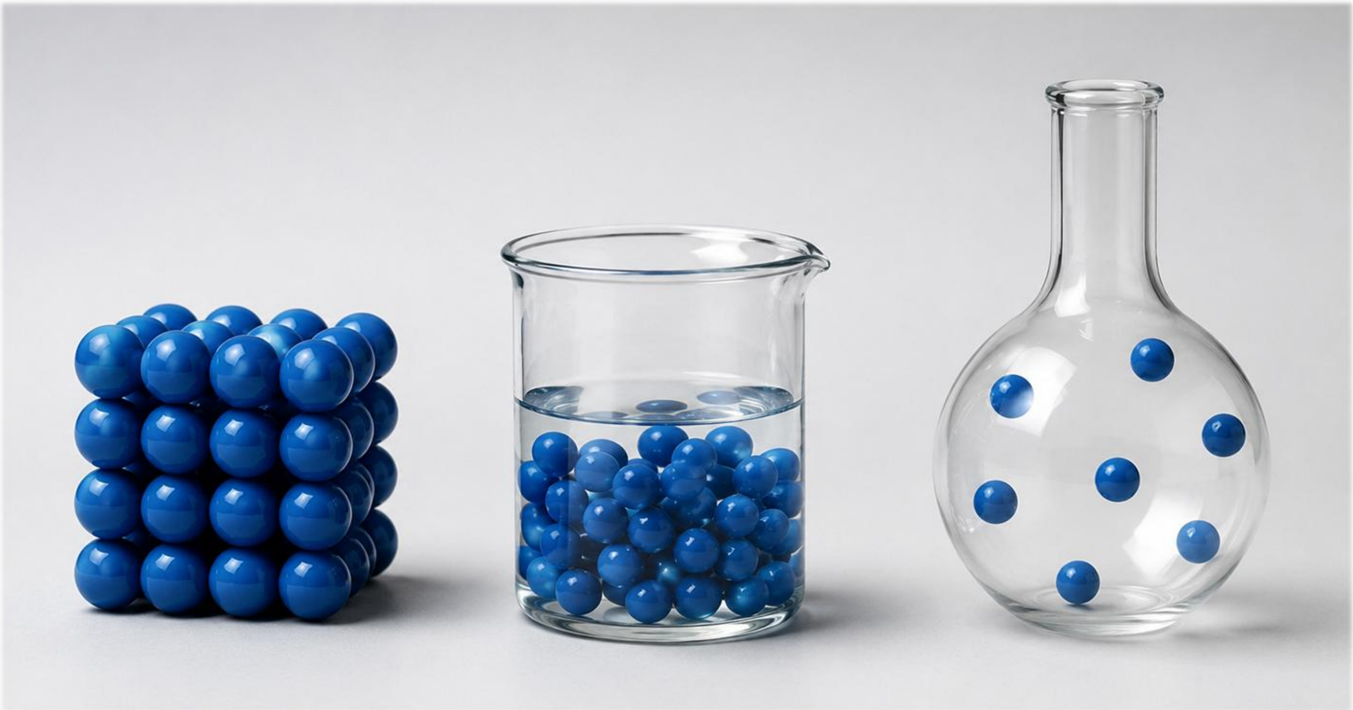

Fig. 2 *Particle models showing the arrangement of particles in a solid (left), liquid (centre) and gas (right). Notice how the spacing and arrangement change across the three states.*

Activity 1.1 Matter Hunt

Look around you. List five things made of matter.

Particle Theory

Scientists explain matter using particle theory.

Particle theory states that all matter is made of tiny particles that are too small to see even with a microscope. These particles are always moving. The way they are arranged and how much energy they have determines whether a substance is a solid, a liquid, or a gas.

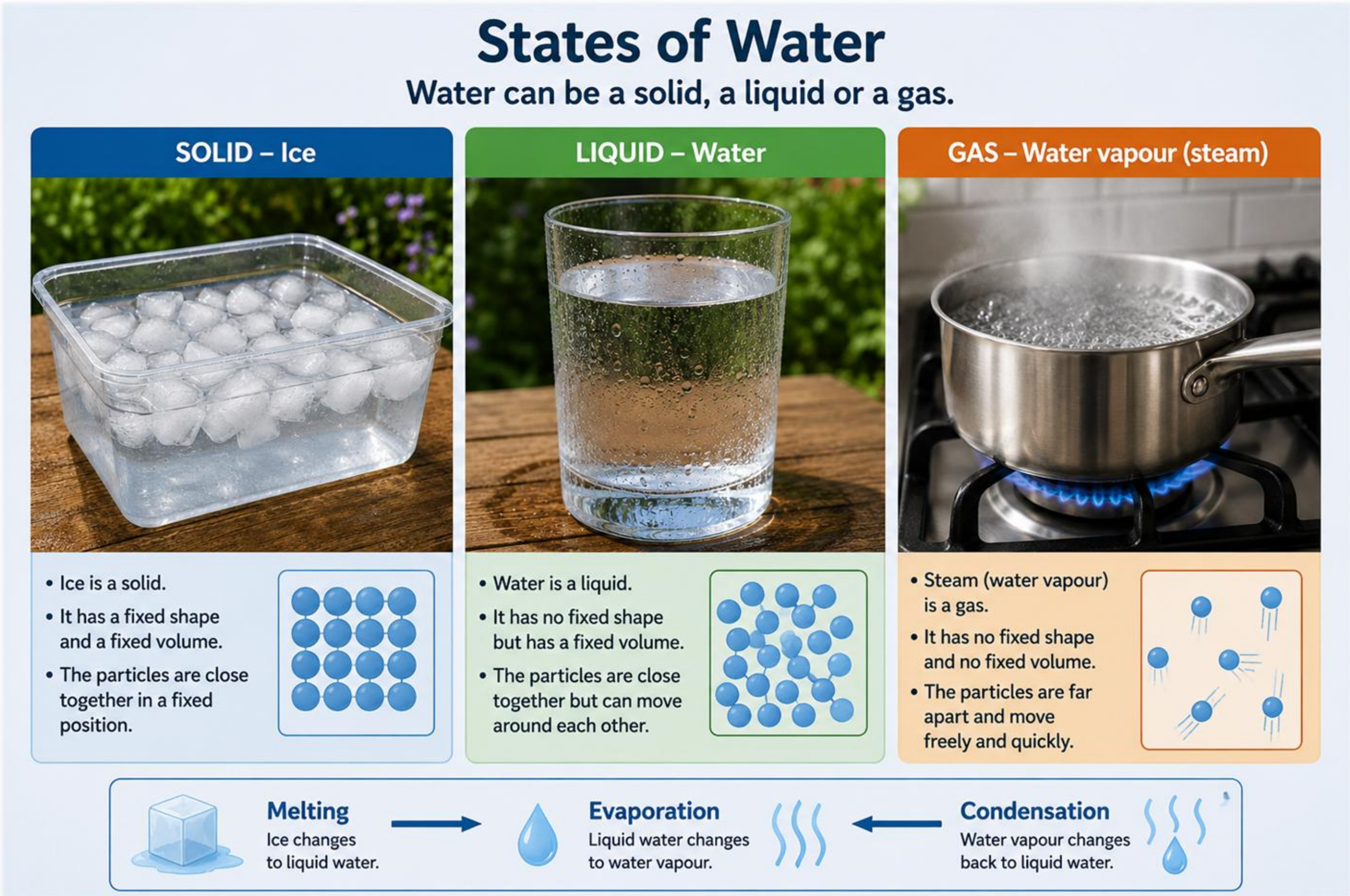

Fig. 3 *States of Water: ice (solid), liquid water, and water vapour (steam). Each panel shows the corresponding particle arrangement and the key properties of that state. The lower panel shows the names of the state changes.*

The Three States of Matter

Matter exists in three main states: solid, liquid, and gas. The difference between them is how the particles are arranged and how much energy they have.

Property	Solid	Liquid	Gas
Shape	Fixed shape	Takes shape of container	Fills entire container
Volume	Fixed volume	Fixed volume	Fills entire container
Particle arrangement	Regular, tightly packed in rows	Random, close together	Random, far apart
Particle movement	Vibrate in fixed positions	Move around each other	Move quickly and freely in all directions
Particle energy	Lowest energy	Intermediate energy	Highest energy
Examples	Ice, wood, metal	Water, milk, oil	Air, steam, oxygen

Changing State

Matter can change from one state to another when it is heated or cooled. When matter gains heat energy, its particles move faster. When it loses heat energy, its particles slow down.

- Heating: particles gain energy, move faster, and the state can change (e.g. ice melts to become liquid water)
- Cooling: particles lose energy, slow down, and the state can change (e.g. water vapour condenses to become liquid water)

Names of State Changes:

Change	Process	What Happens to Particles	Example
Solid → Liquid	Melting	Particles gain energy and break free of fixed positions	Ice melting
Liquid → Solid	Freezing (Solidifying)	Particles lose energy and lock into fixed positions	Water freezing
Liquid → Gas	Evaporating (Vaporising)	Particles gain enough energy to escape the liquid surface	Puddles drying up
Gas → Liquid	Condensing	Particles lose energy and slow down enough to attract each other	Steam on a cold window
Solid → Gas	Subliming	Particles go directly from solid to gas without becoming liquid	Dry ice disappearing

Note: sublimation is extension content. It is not required at KS2 but does appear in KS3 Chemistry.

🌍 CHEMISTRY IN ACTION: Breathing

Every time you breathe in, oxygen gas particles from the air move into your lungs. Your body uses those particles to release energy. When you breathe out, you release carbon dioxide gas particles. Both oxygen and carbon dioxide are gases at room temperature. Their particles are far apart and move freely, which is why they can diffuse easily in and out of your blood.

💡 LOOKING AHEAD TO SECONDARY SCHOOL

In Year 7, you will use the particle model to explain diffusion, gas pressure, and density changes. You will also meet the term vapour, meaning a gas that is below its boiling point. Everything in this section is laying the groundwork for that.

Activity 1.2 Particle Diagrams

Draw how you think particles are arranged in each state of matter. Use Fig. 2 and Fig. 3 above to guide you.

SOLID	LIQUID	GAS
Draw here	Draw here	Draw here

Reflection

Why do solids keep their shape but liquids and gases do not? Use particle theory in your answer.

➨ **Investigation 1 — Ice, Water, Steam:** Complete this investigation (in the Appendix) during or after working through Section 1. It applies everything you have learned here about particle theory and state changes directly to a practical context.

SECTION 2

MIXTURES & SEPARATION

What Is a Mixture?

A mixture is two or more substances combined together.

In a mixture, the substances do not react chemically with each other. They keep their own properties and can usually be separated. Examples: sand and salt, cereal in milk, muddy water, air (a mixture of gases).

Dissolving

When a substance dissolves, its particles spread out evenly through the liquid.

The substance seems to disappear, but it is still there. The individual particles have spread out into the water. The mixture formed when a solid dissolves in a liquid is called a solution. The dissolved substance is the solute. The liquid it dissolves in is the solvent.

Fig. 4 *A solid dissolving in water. As the solid enters the water, its particles spread out and mix between the water particles, forming a solution. The solid appears to vanish, but mass is conserved. Nothing has been destroyed.*

Term	Meaning	Example
Soluble	Can dissolve in a liquid (usually water)	Sugar, salt, copper sulphate
Insoluble	Cannot dissolve in a liquid	Sand, chalk, flour
Solution	A mixture formed when a solute dissolves in a solvent	Salt water, sugar water
Solute	The substance that dissolves	Salt (in salt water)
Solvent	The liquid that does the dissolving	Water (in salt water)

Important: dissolving is a physical change, not a chemical one. The solute can be recovered (e.g. by evaporation). No new substance is formed.

Activity 2.1 Soluble or Insoluble?

Predict whether each substance will dissolve in water. Tick your answer.

Substance	Soluble?	Insoluble?
Sugar		
Sand		
Salt		
Flour		

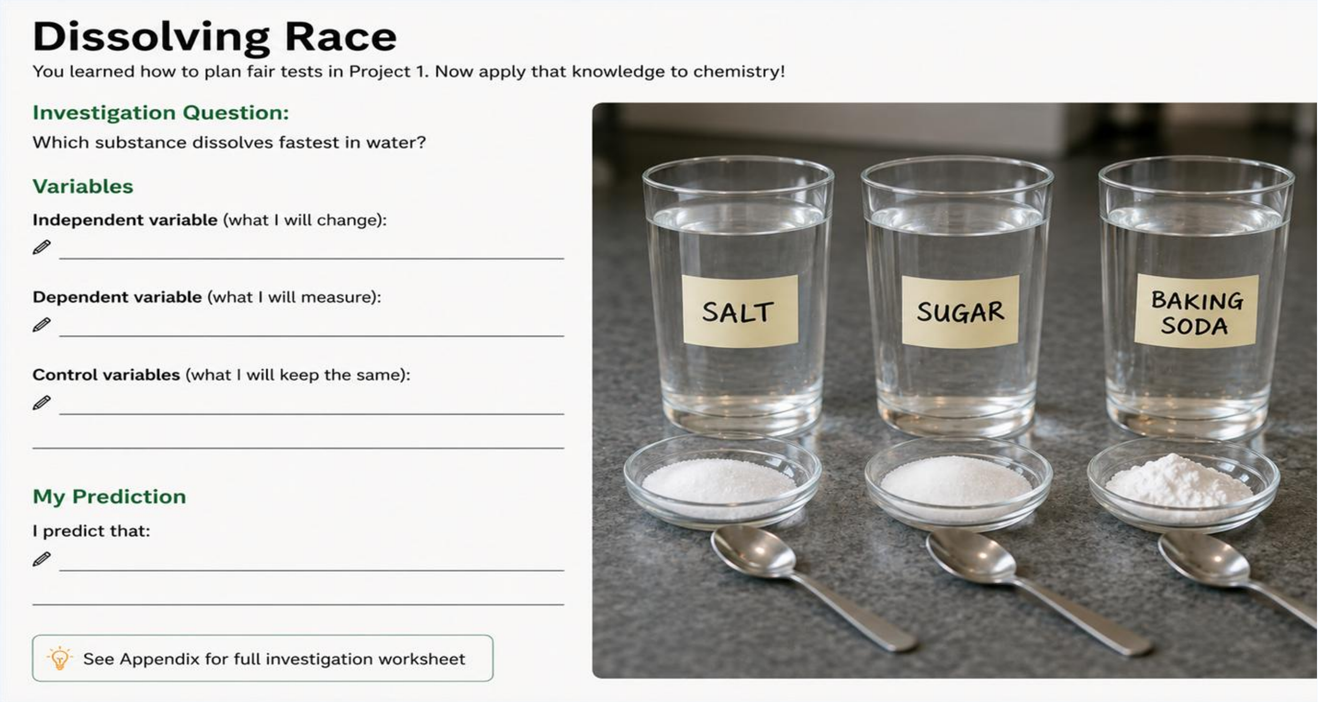

Fig. 5 *Investigation 2: The Dissolving Race. Three glasses contain equal volumes of water, ready to test how quickly salt, sugar, and baking soda dissolve. Each dish below holds the same amount of each substance.*

INVESTIGATION 2 — The Dissolving Race

Investigation Question

Which substance dissolves fastest in water?

Variables

Variable Type	Your Answer
Independent variable (what I will change)	
Dependent variable (what I will measure)	
Control variables (what I will keep the same)	

My Prediction

I predict that:

Separating Mixtures

Chemists use different techniques to separate mixtures. The method you choose depends on the properties of the substances in the mixture.

1. Filtration

Filtration separates an insoluble solid from a liquid by passing the mixture through filter paper. The solid (called the residue) stays on the paper; the liquid (called the filtrate) passes through.

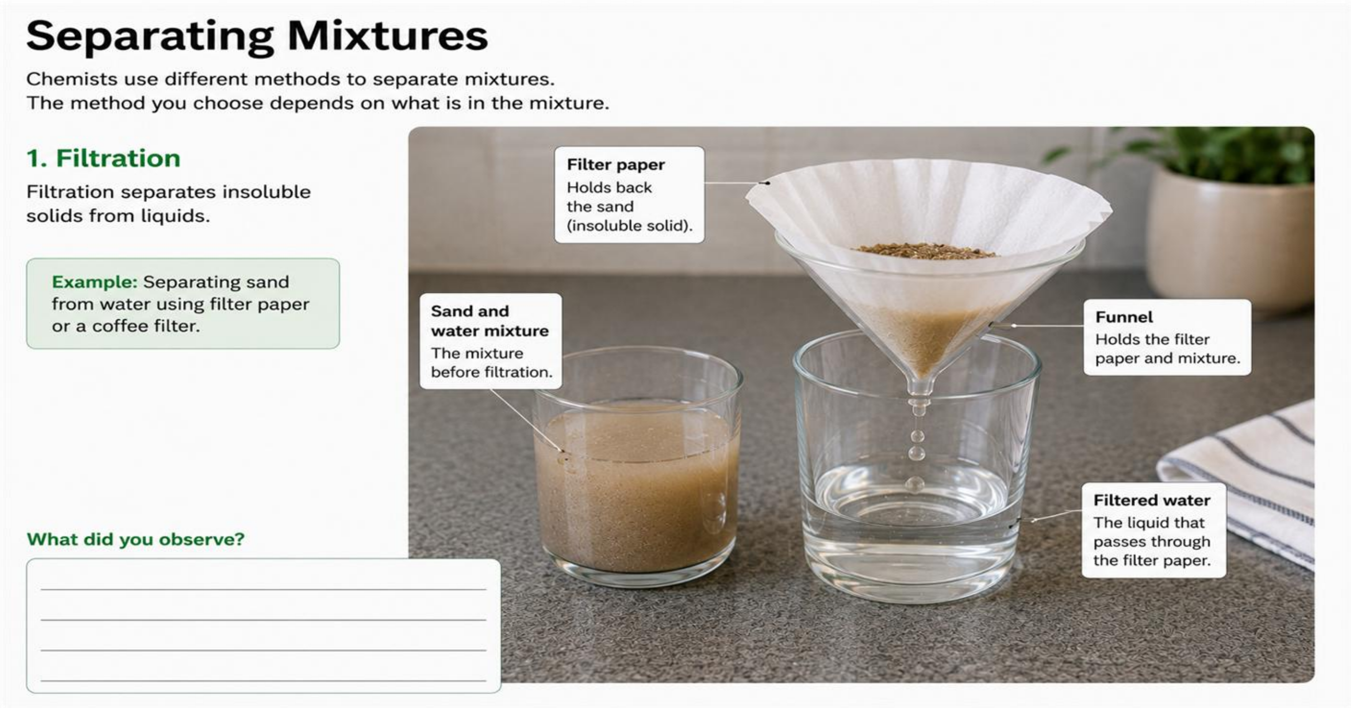

Fig. 6 *Filtration setup. The sandy water mixture (left) is poured through filter paper in a funnel. The sand (insoluble) is trapped as the residue in the filter paper; the clear filtered water (filtrate) collects in the glass below.*

2. Evaporation

Evaporation is used to separate a dissolved solid from a solution. The solution is heated gently; the solvent evaporates as a gas, leaving the solid behind as crystals.

Fig. 7 *Evaporation. A salt solution is heated in an evaporating dish over a Bunsen burner. The water evaporates, leaving white salt crystals visible on the surface. Steam can be seen rising as the water escapes as a gas.*

CHEMISTRY IN ACTION: Salt from Seawater

In hot countries near the sea, seawater is channelled into large shallow pans called salterns or salt pans. The sun's heat evaporates the water; salt crystals are left behind and harvested. This method has been used for thousands of years and still produces a large proportion of the world's salt today.

3. Sieving

Sieving separates solid particles of different sizes. Smaller particles pass through the holes in the sieve; larger particles are retained.

Fig. 8 *Sieving experiment. Fine flour particles pass through the fine mesh of the sieve and collect in the bowl below. Larger particles, such as stones and lumps, are too big to pass through and remain in the sieve. This is a physical separation: no chemical change occurs.*

The key variable in sieving is the aperture size; the diameter of the holes in the mesh. If the aperture is larger than a particle, the particle passes through; if smaller, it is retained. In the laboratory and in industry, scientists and engineers often use a series of sieves stacked on top of each other, each with a different aperture size. This produces several separate size fractions in a single operation rather than just two. Soil scientists use this technique to analyse particle size distribution, which tells them about drainage, fertility, and the history of how the soil was formed. In secondary school you will encounter many separation techniques, and one of the first questions a chemist asks is always: does this technique work because of a physical difference (like size, magnetism, or boiling point) or a chemical one? Sieving is firmly in the physical category.

4. Magnetic Separation

Magnetic separation uses a magnet to remove magnetic materials (such as iron or steel) from a mixture. Non-magnetic materials are unaffected.

Example: removing iron filings from a sand and iron filings mixture using a bar magnet.

Fig. 9 *Magnetic separation in the laboratory. A bar magnet is drawn through a mixture of iron filings and sand in a petri dish. The magnetic iron filings are attracted to the magnet and lift clear of the sand, which remains behind. The separated filings are deposited into a clean collection dish.*

It is important to understand which materials are actually magnetic. Iron, steel, nickel, and cobalt are ferromagnetic, they are strongly attracted to magnets. Many other metals, including aluminium, copper, gold, and silver, are not magnetic at all. This is a common misconception: not all metals are magnetic. A bar magnet will lift iron filings from a mixture with sand, but it will leave aluminium powder completely untouched. In secondary school, you will be expected to recall this distinction and apply it when designing separation procedures.

A practical tip that scientists use: wrap the magnet in a thin piece of cling film or paper before drawing it through the mixture. The iron filings still cling to the magnet through the film, but when you remove the film, the filings fall cleanly away into a collection vessel rather than sticking permanently to the magnet. This small technique makes the separation much cleaner and easier to repeat. It is an example of how scientific method includes not just the principle but the practical refinements that make an experiment reliable.

Fig. 10 *Magnetic separation at industrial scale. A powerful overhead electromagnet at a metal recycling facility lifts ferrous metal fragments from a mixed waste conveyor belt. The same principle applies as in the laboratory — magnetic materials are attracted and lifted away, leaving non-magnetic materials behind.*

Industrial electromagnets are used rather than permanent magnets for one key reason: you can switch them off. A permanent magnet holds its load indefinitely, making controlled release impossible. An electromagnet is a coil of wire carrying an electric current. Switch the current off and the magnetic field collapses instantly, dropping the metal. That on-off control is essential when a machine must lift, move, and release thousands of times an hour. Magnetic separation at this scale only captures ferrous metals (iron and steel). Aluminum and copper require a different technique called eddy current separation, where a spinning magnetic rotor induces electrical currents inside non-ferrous metals, and those currents generate a repelling magnetic force that throws the metal clear. It is worth remembering: magnetism here separates materials that are not themselves magnetic.

5. Chromatography

Paper chromatography separates different soluble coloured substances (pigments) in a mixture. Each pigment travels a different distance through the paper, producing a pattern of separated colours called a chromatogram.

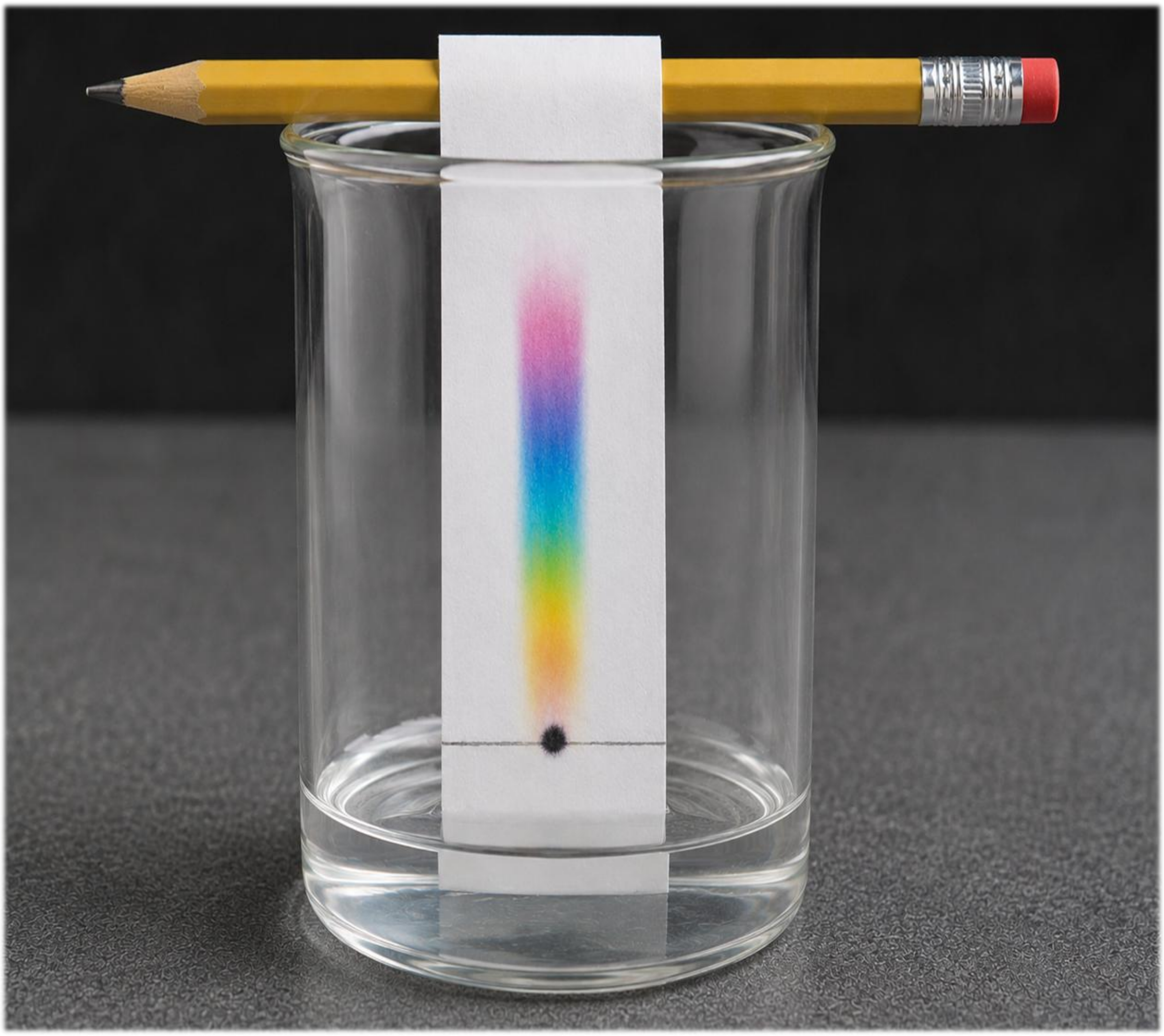

Fig. 11 *Paper chromatography. A single dot of black felt-tip ink was placed at the bottom of a strip of paper. As water moved up the paper, it carried different pigments at different speeds, separating them into distinct coloured bands. Black ink is actually a mixture of several colours.*

CHEMISTRY IN ACTION: Chromatography in Forensics

Forensic scientists use chromatography to examine inks in questioned documents. If a suspect document is written with a different pen from the one claimed, chromatography will reveal this: the ink's chromatogram will show a different pattern of pigments.

6. Crystallisation

Crystallisation is a more controlled form of evaporation. The solution is heated until some solvent evaporates, then left to cool slowly. As it cools, the solute comes out of solution and forms large, regular crystals. Crystallisation produces purer, better-formed crystals than rapid evaporation.

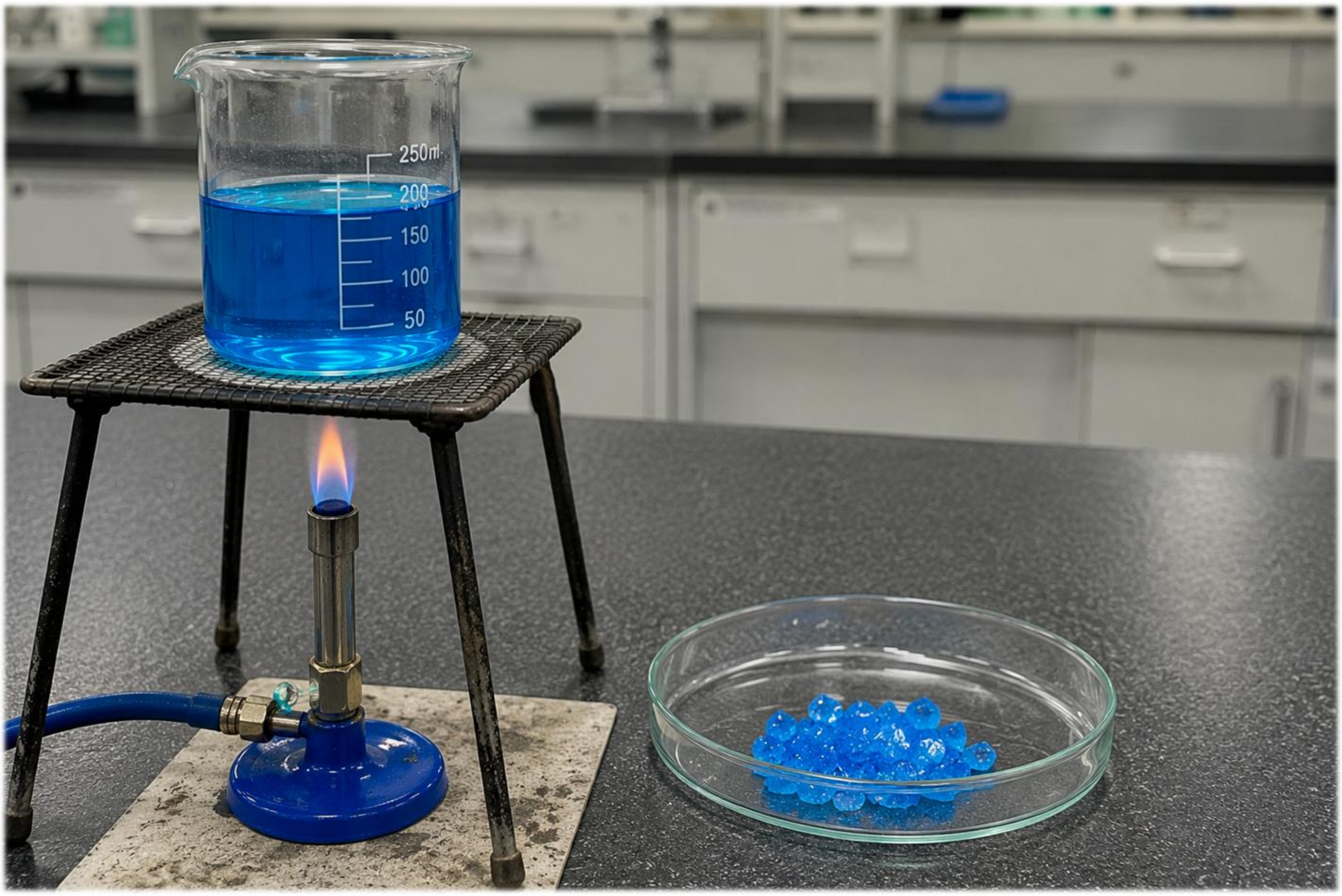

Fig. 12 *Crystallisation. A salt solution is heated in an evaporating dish until partially concentrated, then left to cool slowly. As the solution cools, the dissolved salt comes out of solution and forms large, well-defined crystals. Crystallisation produces purer, more regular crystals than rapid evaporation.*

Cooling rate controls crystal size. Slow cooling gives particles time to settle into their correct lattice positions, producing large, well-formed crystals. Rapid cooling forces many tiny crystals to form at once. Leaving the solution to cool slowly is a deliberate choice that determines product quality.

Crystallisation is also a powerful purification tool. As the lattice forms, impurities are excluded because they do not fit the repeating structure. They remain in the leftover liquid, called the mother liquor, which is drained away. Repeating this process, known as recrystallisation, purifies everything from aspirin to semiconductors.

7. Distillation

Distillation separates a liquid from a solution by boiling it and condensing the vapour. The liquid with the lower boiling point boils first, becomes a vapour, travels through the condenser (where it cools and liquefies), and is collected as a pure liquid called the distillate.

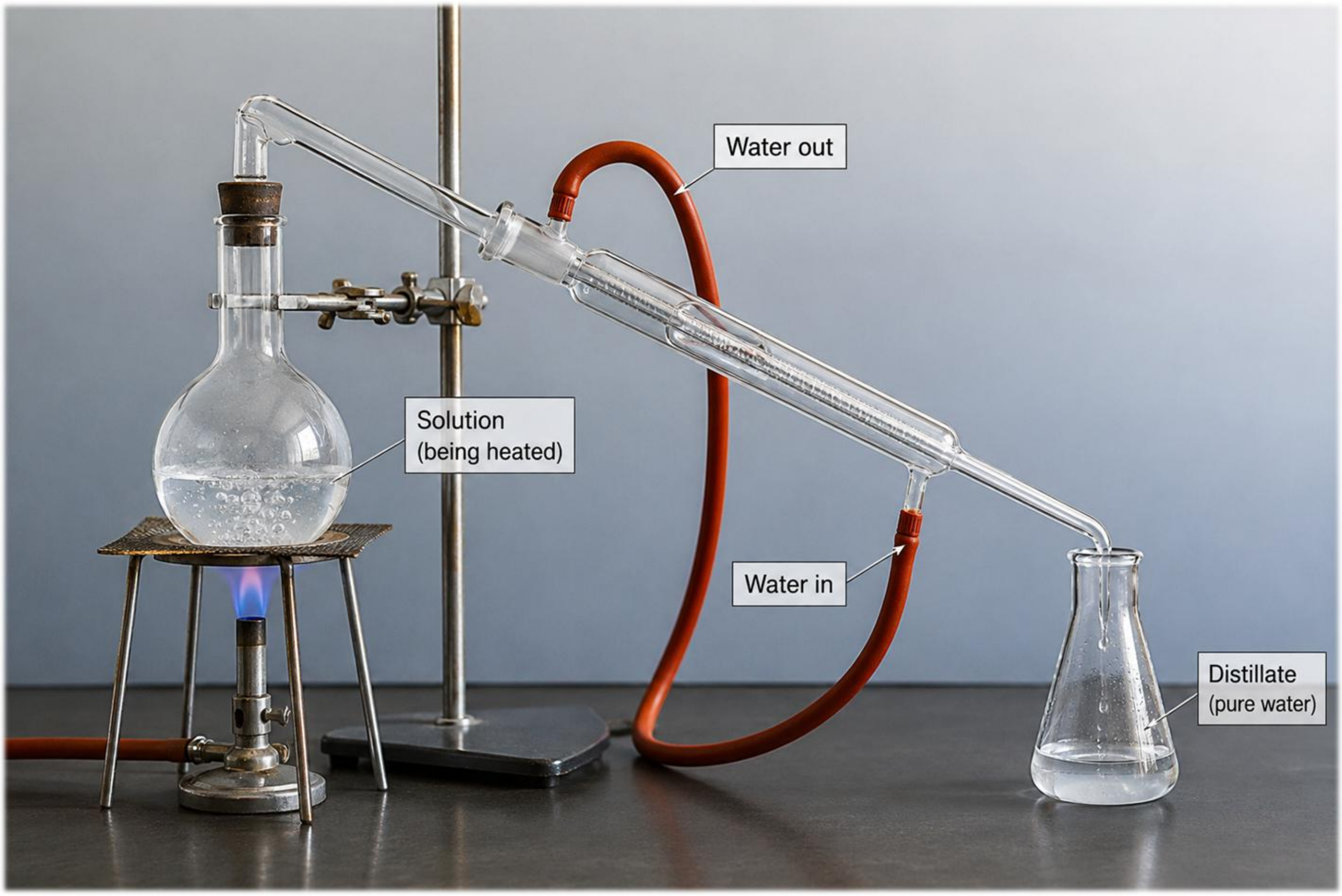

Fig. 13 *Simple distillation apparatus. The salt solution is heated in the round-bottomed flask. Water vapour rises, passes through the condenser (cooled by water flowing through the outer jacket), condenses back to liquid water, and is collected as pure distillate in the conical flask. The salt remains in the original flask.*

CHEMISTRY IN ACTION: Clean Drinking Water

Water treatment plants use a combination of filtration (to remove particles), chemical treatment (to kill bacteria), and sometimes distillation to produce water clean and safe enough to drink. In parts of the world where freshwater is scarce, seawater is purified into drinking water using large-scale distillation plants called desalination plants.

LOOKING AHEAD TO SECONDARY SCHOOL

All seven separation methods described here appear in the KS3 Chemistry curriculum. You will carry out most of them as practical experiments in Year 7 and Year 8. Knowing which method suits which mixture is a skill that comes up in GCSE exams.

Activity 2.2 Which Separation Method?

For each mixture below, choose the best separation method and give your reason.

Mixture	Best Separation Method	Reason
Sand and water		
Salt water (to recover salt)		
Pure water from salt water		
Iron filings and sand		
Different coloured inks		
Gravel and sand		

SECTION 3

PHYSICAL & CHEMICAL CHANGE

Not all changes are the same. Some can be reversed; others cannot. The key question is: has a new substance been formed?

Physical Change	Chemical Change
Only the appearance or state of the substance changes	One or more new substances are formed
No new substance is formed	Also called a chemical reaction
Usually reversible (e.g. melting ice can be refrozen)	Usually irreversible (the original substances cannot easily be recovered)
Examples: melting ice, dissolving sugar, cutting paper, evaporating water	Examples: burning wood, rusting iron, baking a cake, the vinegar and bicarbonate reaction

CHEMISTRY IN ACTION: Baking

When you bake a cake, several chemical reactions happen at once. Baking powder reacts with liquid and acid to produce carbon dioxide gas, which makes the mixture rise. Heat causes proteins in the eggs and flour to change shape permanently, a process called denaturation. New flavour molecules form through the Maillard reaction. A baked cake looks, smells, and tastes completely different from the raw ingredients. You cannot unbake it.

Key Principle — Conservation of Mass

In both physical and chemical changes, **mass is always conserved**. Matter is never created or destroyed. If you measured the total mass of all the reactants before a reaction, and all the products afterwards, both totals would be the same. This is one of the most important laws in science and is a core requirement of the National Curriculum at both KS2 and KS3.

Signs of Chemical Change

You can tell a chemical reaction has probably occurred if you observe one or more of the following. None of these signs on its own is definitive. The key evidence is that a new substance with different properties has been formed.

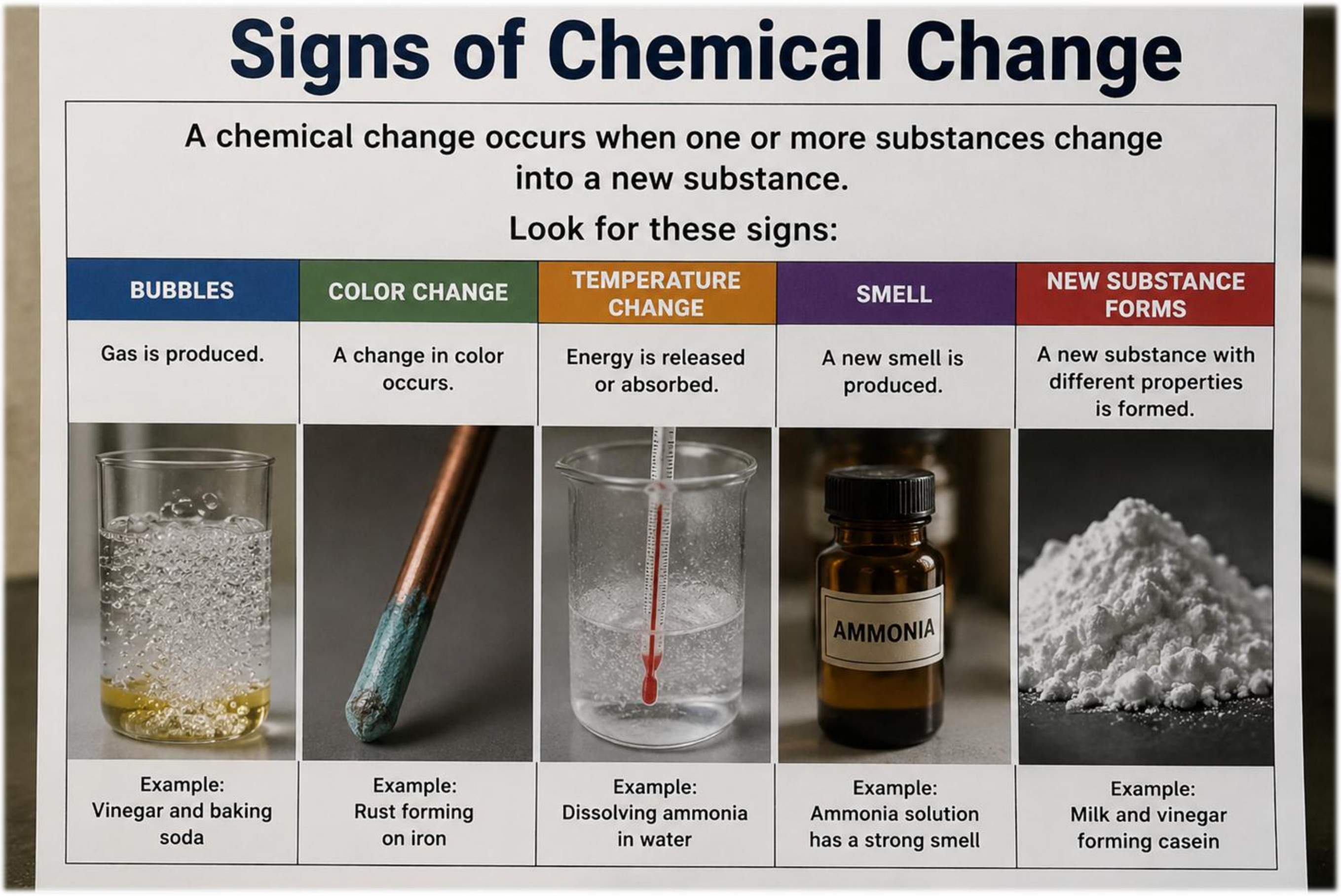

Fig. 14 *Signs of chemical change. Each panel shows one observable sign that a chemical reaction may have occurred: bubbles, gas produced, colour change, temperature change, new smell, and new substance forming. Note: dissolving ammonia in water produces a temperature change but remains a physical change. Always look for multiple signs together.*

Sign	What It Suggests	Example
Bubbles or gas produced	A new gas substance has formed	Vinegar + bicarbonate of soda → CO_2 gas
Colour change	A new substance with different properties is forming	Iron rusting — grey/silver to orange-brown
Temperature change	Energy is released (exothermic) or absorbed (endothermic)	Hand warmers get hot (exothermic)
New smell	New volatile substances have been formed	Eggs cooking; bread baking
New solid forms (precipitate)	A new insoluble substance has formed in solution	Mixing two clear solutions to form a cloudy liquid
Light or flames produced	A rapid, highly exothermic reaction is occurring	Burning magnesium producing white light

Reactants and Products

In a chemical reaction, the substances you start with are called reactants. The new substances formed are called products.

Reactants → Products

Example: the vinegar and bicarbonate reaction:

acetic acid + sodium bicarbonate → carbon dioxide + water + sodium acetate

(vinegar) (baking soda) (the bubbles) (a salt)

LOOKING AHEAD TO SECONDARY SCHOOL

In Year 7, you will write word equations for chemical reactions and eventually symbol equations. You will also study exothermic and endothermic reactions in more depth. You are already thinking about reactants, products, and energy changes.

Activity 3.1 Physical or Chemical?

Read each change below. Decide if it is physical or chemical, and explain how you know.

Change	Physical or Chemical?	How Do You Know?
Ice melting into water		
Burning wood		
Dissolving salt in water		
Baking a cake		
Tearing paper		
Rusting iron nail		

INVESTIGATION 4 — Fizz, Foam & Gas

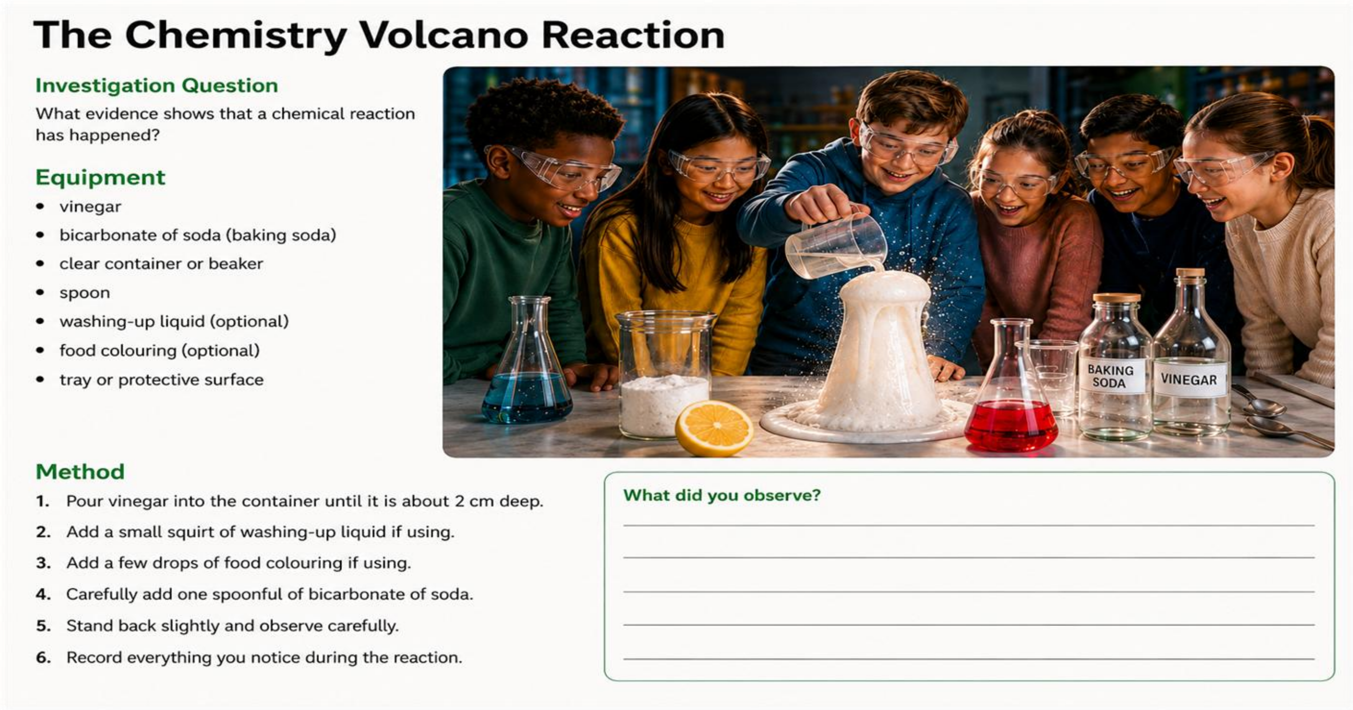

Fig. 15 *Investigation 4: The Chemistry Volcano Reaction. Equipment laid out for the Fizz, Foam and Gas investigation: vinegar, washing-up liquid, food colouring, bicarbonate of soda, and a beaker on a tray. The same reaction produces the dramatic foam eruption shown in Fig. 1.*

Investigation Question

What evidence shows that a chemical reaction has taken place?

Equipment

- Vinegar (acetic acid solution)
- Bicarbonate of soda (sodium bicarbonate / baking soda)
- Clear container or beaker
- Spoon
- Washing-up liquid (optional, makes the CO_2 foam more visible)
- Food colouring (optional)
- Tray or protective surface

Method

1. Pour vinegar into the container until it is about 2 cm deep.
2. Add a small squirt of washing-up liquid if using.

3. Add a few drops of food colouring if using.
4. Carefully add one spoonful of bicarbonate of soda.
5. Stand back slightly and observe carefully.
6. Record everything you notice during the reaction.

Fig. 16 *The Kitchen Volcano in action. Vinegar, washing-up liquid, bicarbonate of soda, and red food colouring produce a vigorous foam eruption. The carbon dioxide gas produced is trapped by the washing-up liquid, creating the dramatic foam. All equipment and the reaction tray are clearly labelled.*

My Prediction

What do you think will happen when the substances are mixed?

Observation Table

Time	What I Saw	What I Heard	Other Observations
At the start			
During the reaction			
After the reaction			

Signs of Chemical Change — Tick what you observed

Sign of Chemical Change	Did You Observe This? (✓ or ✗)
Bubbles forming	
Foam produced	
Gas released	
Temperature change (feel the container)	
New smell (different from vinegar alone)	
Colour change (if food colouring used)	

Conclusion Questions

1. What evidence suggested a chemical reaction happened? List at least two observations.

2. The gas produced is carbon dioxide (CO_2). How does the washing-up liquid make the CO_2 visible?

3. Was this a physical or chemical change? Justify your answer with evidence from your observations.

Evaluation

What worked well in this investigation?

What could be improved?

How could the investigation be made more reliable (i.e. give more trustworthy results)?

> **★ EXTENSION: The Kitchen Volcano**
> How could you make the reaction more dramatic, or change its rate? Change one variable at a time and predict what will happen.
> Try: more bicarbonate of soda / warm vinegar instead of cold / a narrower container / different types of acid (lemon juice instead of vinegar).

Extension Challenge — Think Like a Chemist

What do you think would happen if:

- More bicarbonate of soda was added?

- Warm vinegar was used instead of cold?

- The container was sealed tightly after mixing?

> **➨ Investigation 4 — Fizz, Foam & Gas:** Complete this investigation (in the Appendix) during or after working through Section 3. It applies everything you have learned here about chemical change directly to a practical context.

SECTION 4

KITCHEN ACIDS & ALKALIS

What Are Acids and Alkalis?

Scientists classify substances as acids, alkalis, or neutral. These classifications describe genuine chemical properties that affect how a substance behaves and reacts.

Acids	Alkalis	Neutral
Vinegar (acetic acid solution)	Bicarbonate of soda solution	Water
Lemon juice (citric acid)	Soap	Sugar solution
Orange juice (citric and ascorbic acid)	Baking powder	Salt solution (at pH 7)
Fizzy drinks (carbonic acid)	Toothpaste	

The pH Scale

Scientists use the pH scale to measure how acidic or alkaline a substance is. The pH scale runs from 0 to 14. pH 7 is neutral. Below 7 is acidic (lower = more acidic). Above 7 is alkaline (higher = more alkaline).

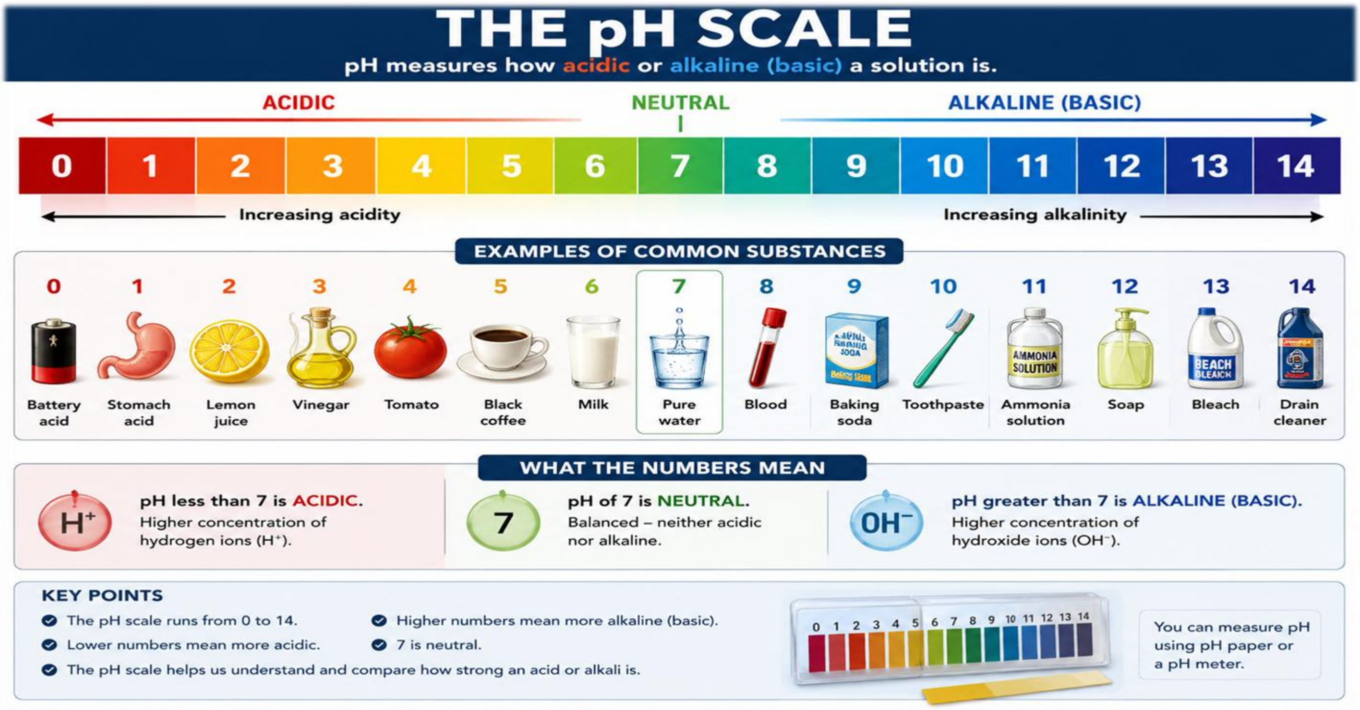

Fig. 17 *The pH scale from 0 to 14, with examples of common substances at each pH value. The scale shows how pH relates to the concentration of hydrogen ions (H+). You will explore this fully in KS3 Chemistry.*

pH Range	Classification	Typical Examples
pH 1 – 3	Strong acid	Battery acid (pH 0–1), stomach acid (pH 1–2), lemon juice (pH 2)
pH 4 – 6	Weak acid	tomato juice (pH 4), black coffee (pH 5)
pH 7	Neutral	Pure water
pH 8 – 10	Weak alkali	Baking soda (pH 8–9), soap (pH 9–10), toothpaste (pH 9)
pH 11 – 14	Strong alkali	Ammonia (pH 11), bleach (pH 13), drain cleaner (pH 14)

Indicators

An indicator is a substance that changes colour to show whether another substance is acidic, neutral, or alkaline. Red cabbage juice contains a natural pigment (anthocyanin) that is an excellent indicator you can make at home.

CHEMISTRY IN ACTION: Indigestion Tablets

Your stomach produces hydrochloric acid to digest food. When too much acid builds up, you feel indigestion. Indigestion tablets contain an alkali, usually calcium carbonate or magnesium hydroxide, which neutralises the excess acid and raises the pH back towards 7. The discomfort fades because the reaction has occurred. Neutralisation is not just a laboratory concept: it happens in your body every day.

Neutralisation

Neutralisation is a chemical reaction between an acid and an alkali. The products are a salt and water, and the pH moves towards 7.

acid + alkali → salt + water

The vinegar and bicarbonate reaction you investigated in Section 3 is an example of neutralisation. Acetic acid (in vinegar) reacts with sodium bicarbonate (an alkali) to produce sodium acetate (a salt), water, and carbon dioxide gas.

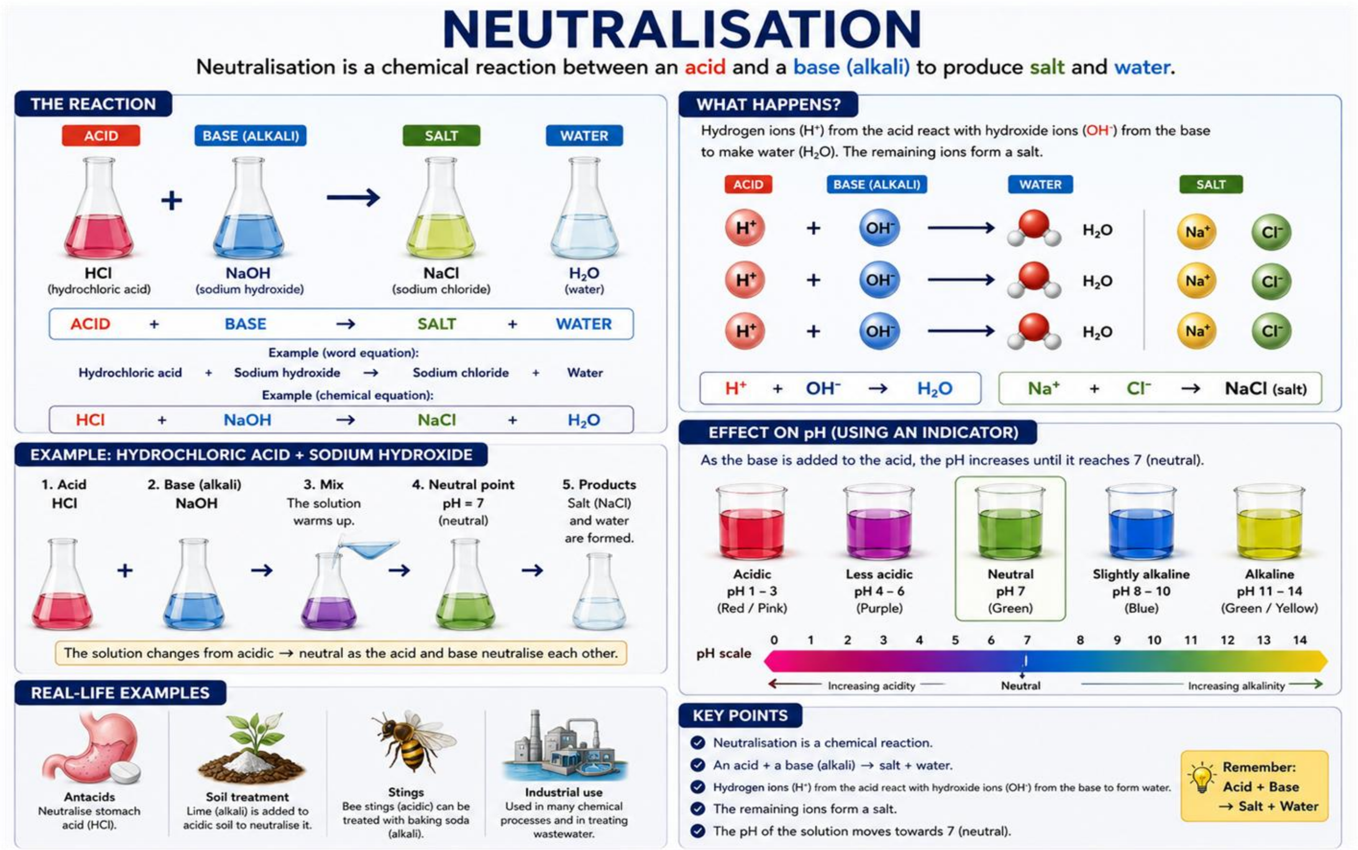

Fig. 18 *Neutralisation diagram. Left panel: the acid-alkali reaction producing salt and water, with the example of HCl + NaOH forming NaCl + H2O. Right panel: the effect on pH as a base is added to an acid, showing the colour changes of a universal indicator. The HCl/NaOH example is GCSE-level extension content. Do not worry about the formulas yet.*

Activity 4.1 Kitchen pH Detective — Predictions

Before carrying out Investigation 4, predict whether each substance is acidic, alkaline, or neutral.

Kitchen Substance	My Prediction (Acid / Alkali / Neutral)	Actual Result
Lemon juice		
Soap		
Tap water		
Vinegar		
Bicarbonate of soda solution		
Orange juice		

INVESTIGATION 5 — Kitchen pH Detective

Investigation Question

Are common kitchen substances acidic, alkaline, or neutral?

Equipment

- Red cabbage (half a head)
- Water and a pan (with adult supervision)
- Strainer
- Small clear containers or test tubes
- Substances to test: vinegar, lemon juice, bicarbonate of soda solution, soap solution, tap water, orange juice

Method — Making the Red Cabbage Indicator

1. Chop the red cabbage into small pieces (adult supervision required).
2. Boil the cabbage in water for 10 minutes (adult supervision required).
3. Strain the liquid carefully into a clean jug. This deep purple liquid is your indicator.
4. Allow the indicator to cool completely before use.
5. Add a small amount of indicator to each container, then add a few drops of each test substance.

Colour Guide — Red Cabbage Indicator

Colour with Indicator	Classification	Approximate pH
Red / Pink	Strong acid	pH 1 – 3
Purple / Mauve	Neutral or weak acid	pH 6 – 7
Blue / Blue-green	Weak alkali	pH 8 – 11
Green / Yellow-green	Strong alkali	pH 12 – 14

Results Table

Substance	Colour with Indicator	Acid, Alkali, or Neutral?	Estimated pH
Vinegar			
Lemon juice			
Bicarbonate of soda solution			
Soap solution			
Tap water			
Orange juice			

Conclusion

Which of the kitchen substances tested were acids?

Which were alkalis?

Did any result surprise you? Explain why, using your knowledge of acids and alkalis.

SECTION 5

CHEMISTRY VOCABULARY

Learning precise scientific vocabulary is essential for communicating clearly in science. It also helps you think more accurately: the right word often carries the right meaning built in.

Write a definition for each word below in your own words. Copying a definition from a book is easy. Writing it yourself shows you understand it.

Matter

Particle

Solid

Liquid

Gas

Mixture

Dissolve

Soluble

Insoluble

Solution

Solute

Solvent

Evaporation

Filtration

Chromatography

Distillation

Physical change

Chemical change

Reactant

Product

Acid

Alkali

Neutral

pH

Indicator

Neutralisation

Reversible

Irreversible

SECTION 6

PREPARING FOR YEAR 7

What to Expect in Secondary School Science

In Year 7, you will work in a science laboratory with specialist equipment. You will follow written methods, use measuring instruments accurately, and record your results in tables.

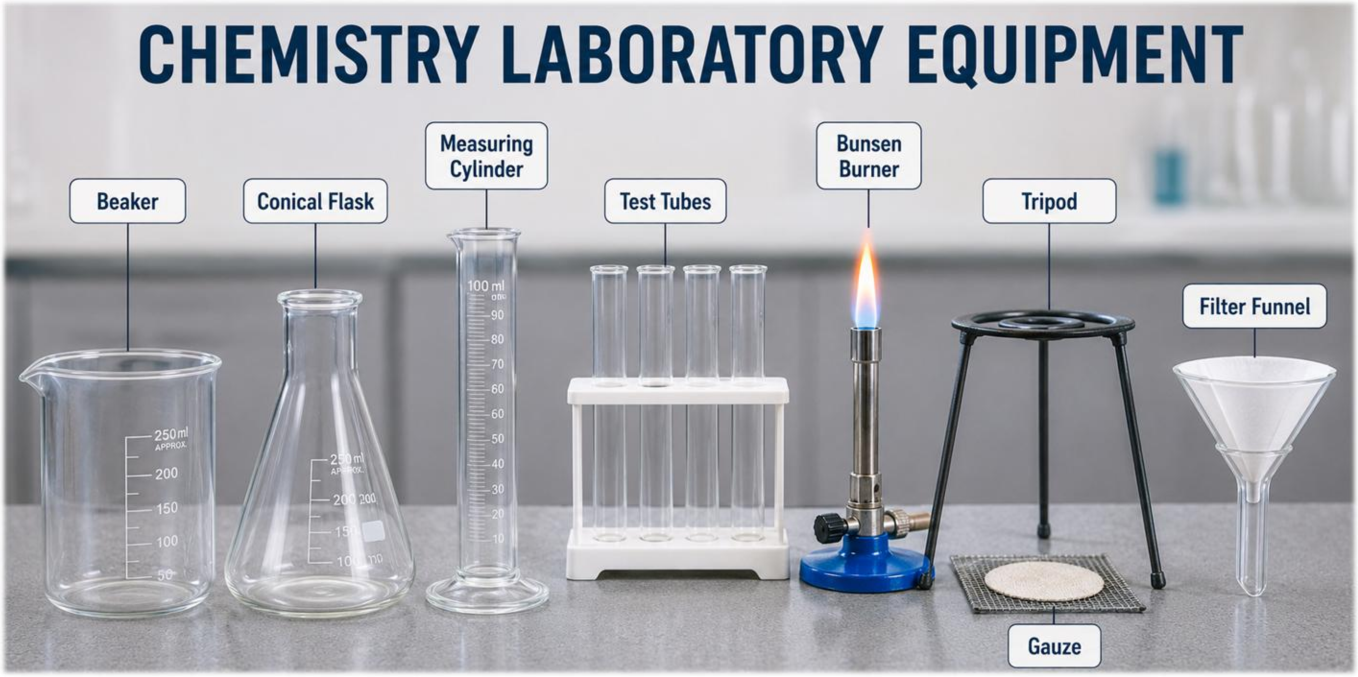

Fig. 19 *Common chemistry laboratory equipment you will use in Year 7. Learn the correct name for each piece. You will be expected to use them accurately in written methods and exam answers.*

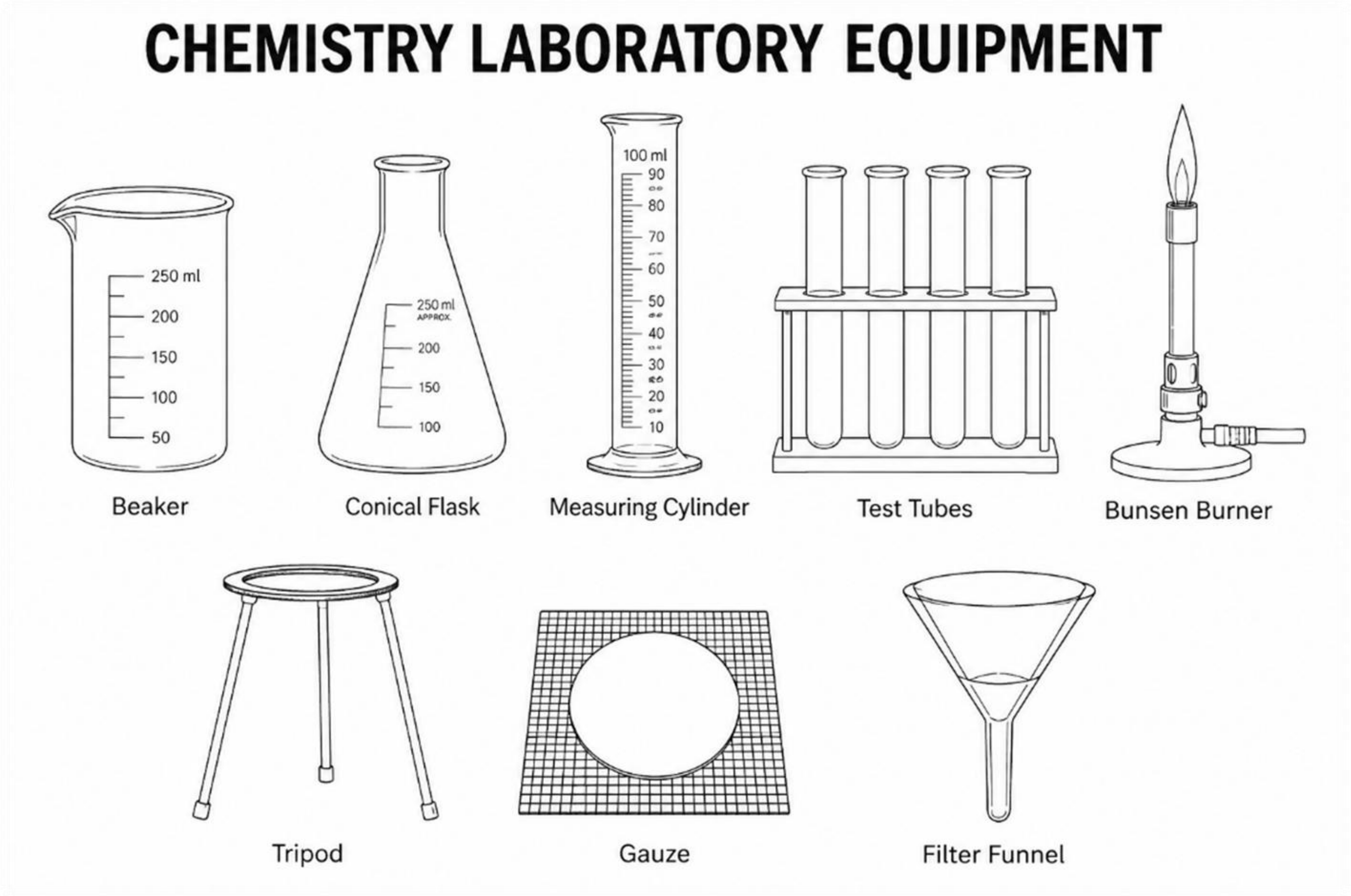

Fig. 20 *Scientific drawings of the same equipment. These are the style of diagram you will draw in secondary school science: plain line drawings, viewed from the side, with ruled label lines. No shading, no 3D perspective.*

Scientific Drawing Skills

In science, we draw equipment in a specific way to show experimental setups clearly and unambiguously.

Rules for good scientific drawings:

- Draw with a ruler and pencil. Use single, clean lines.
- Draw equipment from the side (2D cross-section), not in 3D perspective
- Use ruled straight label lines that end at the part being labelled
- Do not shade or colour the diagram
- Label every important part
- Include a title (e.g. “Simple Distillation Apparatus”)

Safety in the Laboratory

Safety rules in a science laboratory exist because some chemicals and equipment can cause real harm. Your Year 7 teacher will go through all the rules in detail before your first practical lesson. Take them seriously.

Safety Rule	Why It Matters
Wear safety goggles when instructed	Protects your eyes from chemical splashes and debris
Tie long hair back	Prevents hair from catching fire near a Bunsen burner flame
Never eat or drink in the laboratory	Chemicals can contaminate food or drink and cause poisoning
Follow all instructions carefully	Prevents accidents and ensures your results are valid
Tell your teacher immediately if something spills or breaks	Prevents injury to yourself and others; some spills can be hazardous
Wash your hands after practical work	Removes traces of chemicals from your skin

What You Will Study in Year 7 Science

Because you have completed this Kitchen Chemistry project, you will already be familiar with many of the topics in Year 7 Chemistry. Here is what comes next:

- Atoms and elements: the building blocks of all matter. You have studied particles. In Year 7, you will go deeper.
- The Periodic Table: how elements are organised by their properties.
- Compounds and chemical formulae: how atoms combine to form new substances.
- More chemical reactions: word equations, symbol equations, and conservation of mass.
- All seven separation methods as hands-on practical experiments
- More acids and alkalis, including universal indicator and measuring pH precisely
- Safe use of the Bunsen burner

WELL DONE!

By completing this Kitchen Chemistry project, you have covered real ground in KS3 Chemistry before secondary school has even begun. You know particle theory, the three states of matter, seven separation methods, physical and chemical change, the signs of a chemical reaction, and how to use the pH scale.

That is a strong foundation. Use it.

FINAL REFLECTION

BECOMING A CHEMIST

Complete the sentences below:

I learned that matter is made of

I can now explain the difference between a physical and chemical change because

One thing I found surprising about chemistry was

One separation method I would like to try again is

One thing I am proud of in this project is

What Have I Achieved?

A checklist:

☐ I can explain particle theory

☐ I can describe solids, liquids and gases

☐ I can explain dissolving

☐ I can choose a separation method

☐ I can identify chemical change

☐ I can explain acids and alkalis

☐ I can plan a fair investigation

☐ I am ready for Year 7 Chemistry

INSPIRE ACADEMIC™ |
YEAR 6 SCIENCE BRIDGING PROGRAMME

Project 2
Kitchen Chemistry

This Kitchen Chemist Badge is awarded to:

You earn this badge because you can now:

✓ Explain what matter is made of using particle theory
✓ Describe and compare solids, liquids, and gases
✓ Separate mixtures using seven scientific methods
✓ Distinguish between physical and chemical changes with evidence
✓ Use the pH scale and indicators to classify substances

Awarded for careful observation, scientific explanation, and genuine chemical thinking.

APPENDIX

KITCHEN CHEMISTRY INVESTIGATIONS

⚠ SAFETY GUIDELINES — ALWAYS FOLLOW THESE RULES

Always work with an adult supervisor present

Do NOT taste or eat anything used in investigations

Clean all equipment thoroughly before and after use

Wash hands thoroughly after all investigations

Work on a clear, protected surface. Use a tray.

Handle hot liquids with care. Use oven gloves and always wait for liquids to cool before handling.

INVESTIGATION 1 — Ice, Water, Steam

Investigation Question

What happens to the arrangement and movement of particles when ice melts and water boils?

Equipment

- Ice cubes
- Kettle or pan (with adult supervision)
- Thermometer (optional)

Method

1. Observe ice cubes carefully. Draw how you think the particles are arranged. (use the table below)

SOLID (Ice)	LIQUID (Water)	GAS (Steam)
Draw here	Draw here	Draw here

2. Watch ice melting. What do you think happens to the particles as they gain energy?

3. With adult supervision, observe water boiling. What does steam tell you about particle movement?

4. Observe steam condensing on a cold surface. What does this tell you about particle energy?

Conclusion

Use particle theory to explain what happens when ice melts and then the water boils:

INVESTIGATION 2 — The Dissolving Race

Investigation Question

Which substance dissolves fastest in water: salt, sugar, or flour?

Equipment

- 3 clear containers (same size and shape)
- Sugar, salt, flour (equal masses of each)
- Warm water at the same temperature for all three
- Spoon, timer, ruler (to measure same depth of water)

Method

1. Pour the same volume of warm water into each container (measure with a ruler).
2. Add the same amount (same number of spoonfuls) of sugar to container 1.
3. Add the same amount of salt to container 2, and flour to container 3.
4. Start the timer. Stir each the same number of times with the same spoon.
5. Record the time taken for each substance to dissolve completely (or note if it does not dissolve).

Results

Substance	Time to Dissolve (seconds)	Observations
Sugar		
Salt		
Flour (control — insoluble)		

Conclusion

Which substance dissolved fastest?

Why do you think this happened? Use particle theory in your answer.

INVESTIGATION 3 — The Kitchen Separation Challenge

The Challenge

You have a mixture of: sand, salt, paper clips, and water. Your task is to separate all four substances and end up with each one on its own.

Available Methods

- Filtration
- Evaporation
- Magnetic separation
- Sieving

Planning

Plan your sequence of steps carefully before you begin. Think: which substances can each method separate?

Step	Method I Will Use	This Will Separate	Reason
Step 1			
Step 2			
Step 3			
Step 4			

Results and Reflection

Did you successfully separate all four substances?

Which step was most difficult? Why?

INVESTIGATION 4 — Fizz, Foam and Gas

Investigation Question

What evidence shows that a chemical reaction has taken place between vinegar and bicarbonate of soda?

Equipment

- Vinegar
- Bicarbonate of soda (baking soda)
- Clear container or beaker
- Spoon
- Tray or protective surface
- Washing-up liquid (optional, makes the CO_2 foam more visible)
- Food colouring (optional)
- Thermometer (optional, to check for temperature change)

Safety

⚠ Before you begin
Work on a tray or protected surface.
Do not taste or eat anything used in this investigation.
Wash your hands thoroughly after finishing.

My Prediction

What do you think will happen when the vinegar and bicarbonate of soda are mixed? Write your prediction and give a reason.

Method

1. Pour vinegar into the container until it is about 2 cm deep.
2. If using, add a small squirt of washing-up liquid and a few drops of food colouring.
3. Carefully add one level spoonful of bicarbonate of soda.

4. Stand back slightly and observe carefully.
5. Feel the outside of the container. Note whether it feels warmer or cooler than before.
6. Record everything you observe in the table below.

Observation Table

Time	What I Saw	What I Heard	What I Felt (temperature)
At the start			
During the reaction			
After the reaction			

Signs of Chemical Change — Tick what you observed

Sign of Chemical Change	Did You Observe This? (✓ or ✗)
Bubbles forming	
Foam produced	
Gas released (CO_2)	
Temperature change (container felt cooler)	
New smell (different from vinegar alone)	
Change in appearance of the mixture	

Conclusion

1. List two pieces of evidence from your observations that show a chemical reaction took place.

2. The gas produced in this reaction is carbon dioxide (CO_2). How does the washing-up liquid make this visible?

3. This reaction felt slightly cooler than room temperature. What does this tell you about the energy involved?

4. Was this a physical or chemical change? Give two reasons for your answer.

Evaluation

What worked well in this investigation?

What could be improved to make the results more reliable?

How could you change one variable to test a new question? Write a new investigation question you could explore.

★ Extension Challenge

Change one variable at a time and predict what will happen:
Try warm vinegar instead of cold. Does the reaction happen faster? Why?
Try more bicarbonate of soda. Does more gas form?
Try lemon juice instead of vinegar. Does an acid-alkali reaction still occur?
Record your prediction, carry out the test, and write what you found.

KITCHEN CHEMISTRY PORTFOLIO GUIDE

Your portfolio shows all the chemistry thinking and investigation work you have done. Include the following in each section:

Section	What to Include
Section 1: States of Matter	Investigation 1 write-up, particle diagrams for solid/liquid/gas with labels, any photos (optional)
Section 2: Dissolving	Investigation 2 results table and conclusion; your explanation of dissolving using particle theory
Section 3: Separation	Investigation 3 planning table, your separation sequence, observations from each step
Section 4: Chemical Changes	Investigation 4 (Fizz & Foam) observation table, signs of reaction ticked, conclusion questions answered
Section 5: Acids & Alkalis	Investigation 5 results table, colours recorded, conclusion; explanation of any surprising results
Final Reflection	Completed reflection sentences; what you are most proud of; how you feel ready for Year 7

INSPIRE Academic™

Learning How Science Works

By E. K. Appiah

Chemistry is not magic.

It is the careful study of matter, change,

and the hidden patterns of the world around us.

In Project 2, pupils learned to:

- Explain what matter is made of using particle theory
- Describe and compare solids, liquids, and gases
- Separate mixtures using seven scientific methods
- Distinguish between physical and chemical changes using evidence
- Use indicators and the pH scale to classify substances
- Plan and evaluate investigations, and record conclusions scientifically

Rigour • Understanding • Purpose

A Message to

Young Scientists

Science begins with curiosity.

Every great discovery started when someone noticed something others overlooked and asked: ***"Why?"***

Throughout this project you have ***observed*** carefully, ***investigated*** thoughtfully, and learned to ***explain*** the world around you using evidence and reason.

You have explored matter, mixtures, particles, reactions, and change.

More importantly, you have begun to develop the habits of a scientist:

CO_2

H_2CO_3

H_2O_2

Dioxide

Curiosity Today Changes Tomorrow

CURIOSITY | CAREFUL OBSERVATION | THOUGHTFUL QUESTIONING | CLEAR EXPLANATION | PERSEVERANCE

These habits will serve you far beyond the science classroom.

THE PURPOSE OF THIS PROGRAMME

The Inspire Year 6 Science Bridging Programme exists to help young people move into secondary school with confidence, competence, and curiosity.

Our goal is not simply to help pupils learn science. Our goal is to help them become the kind of people who can:

- think clearly
- solve problems
- investigate carefully
- understand deeply
- contribute meaningfully to the world around them

WHAT WE BELIEVE

We believe that every child deserves the opportunity to experience:

WONDER in discovery

CONFIDENCE in learning

EXCELLENCE in understanding

HOPE for the future

☆ THE JOURNEY SO FAR ☆

PROJECT 1
THE GREAT SCIENCE INVESTIGATOR
Learning how scientists investigate and discover.

PROJECT 2
KITCHEN CHEMISTRY
Matter, mixtures, particles, and change.
YOU ARE HERE!

PROJECT 3
INVISIBLE WORLDS
Electricity, energy, systems, and connections.

PROJECT 4
LIVING SYSTEMS – BIOLOGY
Cells, systems, life, and the world of living things.

> **"THE FUTURE BELONGS TO THE CURIOUS."**

INSPIRE | SCIENCE

YEAR 6 SCIENCE BRIDGING PROGRAMME

RIGOR • UNDERSTANDING • PURPOSE

Inspire Academic™
Helping young people move from curiosity to confidence.

Visit us at:
inspireacademic.org

Inspire Science